

N CHARITA JAIN

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SUMMARY

Final-year Electronics and Communication Engineering student with hands-on experience in VLSI, embedded systems, IoT, and PCB design. Skilled in schematic design, PCB layout, sensor interfacing, and hardware testing using KiCad, Arduino, and ESP32, with a strong interest in core electronics and hardware development.

EDUCATION

Bachelor of Engineering (BE) -Electronics and communication P A College of Engineering, Mangaluru, Karnataka	CGPA=8.6	Nov 2022-May 2026
Pre-University Course(PUC) -science stream KPCL pu College Jog falls, Shivamogga, Karnataka		June 2020-March 2022

SKILLS

- Circuit Design and Analysis: Multisim, schematic and layout design using Cadence Virtuoso, KiCad.
- Programming Languages: C.
- Software Tools: MATLAB, Cadence tools (Virtuoso,Genus,NCLaunch),KiCad (Schematic,Footprint assignment, PCB Layout).
- VLSI Design & Simulation: CMOS logic design, layout design, DRC/LVS checks.

INTERNSHIP

Role: IoT & Embedded Systems Intern

IIMSTC | Feb 2026-May 2026

- Worked on PCB design using KiCad including schematic design, footprint assignment, PCB routing, and Gerber generation.
- Performed component selection and datasheet analysis for embedded and power electronics circuits.
- Developed embedded and IoT-based projects using Arduino and ESP32 for sensor interfacing and real-time monitoring.

Key Internship Projects:

1.RC Delay LED Control Circuit :

Designed an RC delay LED control circuit using a BC547 transistor, resistor, capacitor, and LED to create a ~2-second delay using RC charging. Performed timing calculations and designed the schematic and PCB layout in KiCad with proper routing and grounding.

2.Smart Blind Cane for Visually Impaired :

Role: Hardware & PCB Team Member

Developed an IoT-based smart walking cane using ESP32 for assisting visually impaired individuals through real-time obstacle and hazard detection. Integrated ultrasonic, IR, water, and accelerometer sensors with GPS and Firebase-based location tracking to provide buzzer/vibration alerts, fall detection, and emergency monitoring for improved safety and independent navigation.

PROJECTS

• Inverter-Based OTA and Design of Gm-C Filter:

Role: Leader

Designed and implemented a 0.6 V inverter-based Operational Transconductance Amplifier (OTA) and a 2nd-order Gm-C low-pass filter using Cadence Virtuoso on a Red Hat platform with 180 nm CMOS technology. The OTA converts voltage to current with tunable transconductance controlled by a bias voltage, while the filter uses the OTA and capacitor to realize a tunable low-pass response. Achieved stable operation, proper biasing, and low-power analog performance through DC, AC, and transient simulations.

• Gas leakage detection :

Role: Member

Developed a gas leakage detection system using an MQ-series gas sensor and Arduino Uno, achieving around 95% detection accuracy for flammable gases. The system triggered alarms and activated audio indicators upon detecting gas concentrations above safe limits, providing an efficient and cost-effective solution for early warning in residential and industrial spaces.

• Ultrasonic sound radar:

Role: Member

Designed and implemented a radar system using an ultrasonic sensor and Arduino Uno, achieving approximately 90% accuracy in object detection within a range of 2 to 100 cm. The radar efficiently detected objects within a 180-degree span, demonstrating high reliability for short-range applications, such as obstacle detection and mapping.

• Real-time VOC detection in fruits and vegetables:

Role: Member

Developed a real-time VOC detection system for fruits and vegetables using Arduino and an MQ-3 sensor. The system measured VOC levels in ppm, detecting ethanol concentrations between 200–500 ppm to indicate ripening. The data was displayed on an LCD screen and in the Blynk app via ESP32, while a buzzer was triggered when VOC levels exceeded a set threshold, providing early spoilage alerts.

CERTIFICATIONS

- Signal Processing using AI - Abeyaantrix soft lab
- VLSI CAD Part I:Logic - Illions
- Microcrite creation - Infosys
- Postman Classroom Program - Postman
Earned a certificate and badge from Postman and COSH NITK Surathkal for completing the program on APIs
- Python 3.4.3 Training - IIT Bombay

PAPER PRESENTED

- “Real-Time Detection of VOCs in Fruits and Vegetables” — Presented at SEMI-COMM TECH SUMMIT: International Conference on Emerging Trends in Electronics & Communication (ICEST-2025), P. A. College of Engineering, Mangaluru, 7–8 May 2025.
- Presented the paper “Inverter-Based OTA and Design of Gm-C Filter” at SEMI-COMM TECH SUMMIT, International Conference on Emerging Trends in Electronics and Communication (ICEST 26), International Conclave on Engineering Science & Technology - 2026, held at P.A. College of Engineering, Mangaluru, on 6 May 2026.

COMPETITIONS/ACHIEVEMENTS/AWARDS

- Awarded Top Academic Performer for outstanding academic performance during the 3rd year of B.E. in Electronics and Communication Engineering.
- Secured 2nd Place in “KAUSHAL 25” Project Competition and Expo organised by ISTE in association with IET, IEEE, ASME-PACE chapter
- Presented our project “Real time detection of VOC in fruits and vegetables” at IEEE Project Expo-2025 at Navkis College of Engineering, Hassan
- Participated in HackSummit 2025 Hackathon at P A College of Engineering, Mangaluru
- Participated in “VLSI 1.0” a three-day workshop by dept.of ECE in association with IME club

WORKSHOP & LEADERSHIP ACTIVITIES

- Resource Person and Coordinator for the hands-on session on CMOS inverter design using Cadence Virtuoso.
- Member of the club IME (Ignited Minds of Electronics)
- Coordinator - Freq-Fusion Hackathon on “Signal Processing using AI” under IME club
- Coordinator for the workshop “Code & Create”
- Coordinator for the workshop “ELECTRONIX”

LANGUAGES

- English
- Kannada
- Hindi

HOBBIES

- Painting/Art
 - Traveling and nature photography
 - Listening to music
 - Exploring new technologies
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